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PROJECT ON

Customer Churn Analysis Dashboard

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Batch No : 520

About me

- **Background : Bachelor of Technology (B.Tech) in Computer Science Engineering(CSE)** from Mahatma Gandhi Institute of Technology(MGIT)
- **Why you want to learn Data Analyst:** I want to learn Data Analytics because it enables me to transform data into meaningful insights that support better business decisions. I am interested in analyzing data, identifying patterns, and solving real-world problems. My goal is to build a successful career as a Data Analyst and continuously enhance my analytical and technical skills.
- **work experience :** I have 2 years of Professional experience in LTM
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Introduction

Customer churn is an important challenge in the telecom industry because losing customers can reduce recurring revenue and affect long-term business growth. Customers have different characteristics, service preferences, contract types, and spending patterns, which can influence their overall experience.

In this project, I have used Power BI to analyze customer demographics, tenure, services, contracts, payment methods, monthly charges, and churn behavior. The analysis helps identify important patterns within the customer base and provides a clear view of factors associated with customer loss.

Business Problem Statement

The telecom company has customers who have discontinued their services, but the level of customer loss varies across different groups. The business needs to understand which customer characteristics, services, contracts, payment methods, and tenure levels are associated with higher churn. The analysis also focuses on identifying where customer loss is concentrated and whether these patterns indicate potential revenue exposure for the company.

The main goal of this project is to identify the customer groups and factors associated with higher churn. The insights can help the company understand customer behavior, focus on areas that need attention, and make better data-driven decisions. Ultimately, the analysis aims to support effective customer retention and reduce potential revenue loss.

Objective of the Project

The key objectives of this project are to:

- Analyze customer demographics and understand the overall customer profile.
- Examine tenure, service subscriptions, internet types, contracts, and payment methods to identify important customer patterns.
- Compare churn across different customer groups and service categories.
- Analyze monthly charges and other key customer metrics to understand their relationship with customer behavior.
- Use interactive Power BI dashboards and filters to explore the data and identify meaningful insights.
- Present the findings in a clear and interactive format to support data-driven business decisions.

Dataset Overview and Data Source

- Dataset: Telecom Customer Churn Dataset
- Records: 7,043 customers
- Features: 21 customer, service, contract, and billing attributes
- Target Variable: Customer Churn (Yes/No)
- Data Source: Telco Customer Churn dataset
- Purpose: To analyze customer behavior and identify factors influencing churn.

Business Requirements / Key Analysis Questions

- 1) What is the proportion of male and female customers?
- 2) How many senior citizens have churned compared to those who haven't?
- 3) How are customers distributed across different tenure ranges?
- 4) How many customers are subscribed to phone service, and how many are not?
- 5) Is there a correlation between the number of subscribed services and the likelihood of churn?
- 6) Which add-on services are most commonly used by customers who churned?
- 7) How does churn differ between Fiber, DSL and customers without internet?
- 8) Does the average monthly charge increase with higher tenure?
- 9) What are the key customer metrics with interactive filters?
- 10) How does churn vary across different contract types?

Data Understanding & Data Preparation

- Examined customer demographics, services, contracts, billing, and churn information.
- Checked the dataset for missing values, duplicates, and incorrect data types.
- Cleaned and transformed the data to make it suitable for analysis.
- Created required calculated columns and DAX measures for KPIs and analysis.
- Prepared the final dataset for Power BI visualization and dashboard development.

Data Cleaning & Transformation using Power Query

- Imported the raw Telco Customer Churn CSV into Power Query.
- Checked all columns for missing/null values and inconsistent entries.
- Converted TotalCharges from text to numeric format and handled its blank values appropriately.
- Verified that Churn contains consistent Yes/No values and checked categorical fields for consistency.
- Corrected data types for numerical fields such as Tenure, MonthlyCharges, and TotalCharges.
- Created Tenure Range to group customers based on their tenure.
- Created Service Count to identify the number of subscribed services for each customer.
- Performed a final quality check for duplicates, nulls, data types, and inconsistent categories before loading the cleaned data into Power BI.

Data Model & Relationships

Data Model & Relationships

- Used a single-table data model for the Telco Customer Churn dataset.
- The table contains customer, service, contract, billing, and churn attributes.
- Since all data is in one table, no table relationships were required.
- Customer ID acts as the unique identifier for each customer.

Data Model Approach

- Followed a simple and efficient single-table approach.
- Used existing fields for analysis and created DAX measures for dynamic calculations.
- The model was designed to support KPIs, charts, slicers, and interactive filtering.

DAX Measures & Calculated Columns

Created calculated columns for customer-level analysis and DAX measures for dynamic KPIs, churn analysis, service adoption, and revenue impact.

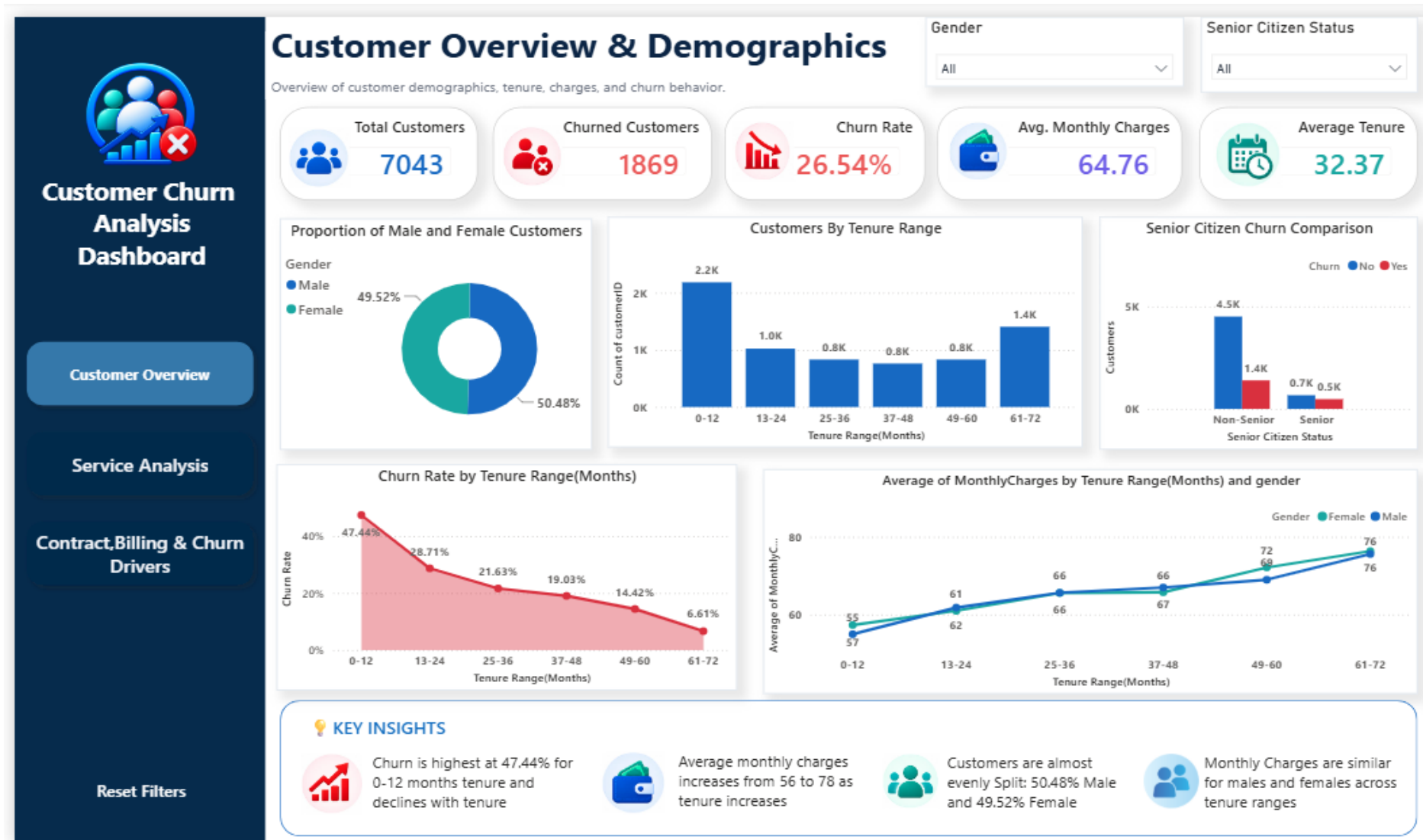
DAX Measures

- Total Customers – Calculates the total number of customers.
- Churned Customers – Calculates the number of customers who have churned.
- Churn Rate – Measures the percentage of customers who have churned.
- Average Tenure – Calculates the average customer tenure in months.
- Average Monthly Charges – Calculates the average monthly charge per customer.
- Lost Revenue – Calculates total revenue associated with churned customers.
- Retained Revenue – Calculates total revenue associated with retained customers.

Calculated Columns

- Tenure Range – Groups customers into meaningful tenure bands for customer lifecycle analysis.
- Service Count – Counts the number of subscribed services/add-ons for each customer.

Dashboard Design & Visualizations

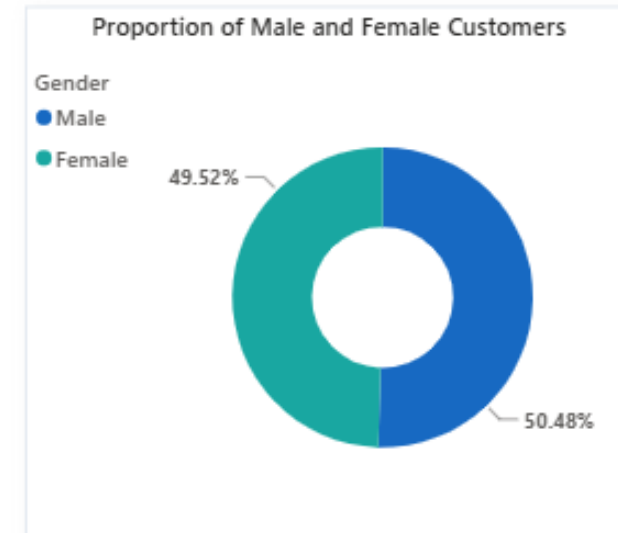


Key Business Insights & Findings

1. What is the proportion of male and female customers?

Key Insight:

- The customer base is almost equally divided between male and female customers.
- Neither gender represents a significantly larger share of the overall customer population.
- This indicates that the company has a fairly balanced customer mix.
- Therefore, gender does not appear to be a major differentiating factor in the overall customer distribution.

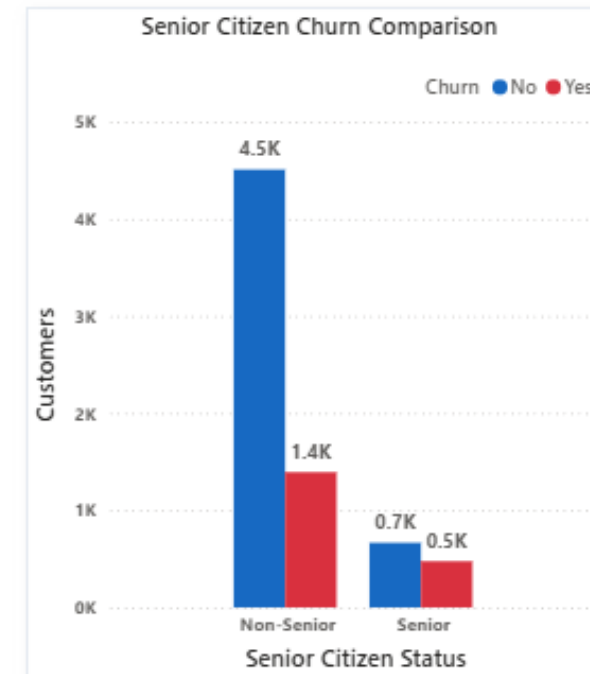


Key Business Insights & Findings

2. How many senior citizens have churned compared to those who haven't?

Key Insight:

- Among senior citizen customers, 476 have churned while 666 have remained with the company.
- Although the number of retained senior customers is higher, the churn rate for senior citizens is noticeably higher than for non-senior customers.
- This indicates that senior customers form a comparatively higher-risk segment.
- The difference makes senior-citizen status worth considering when analyzing customer churn.

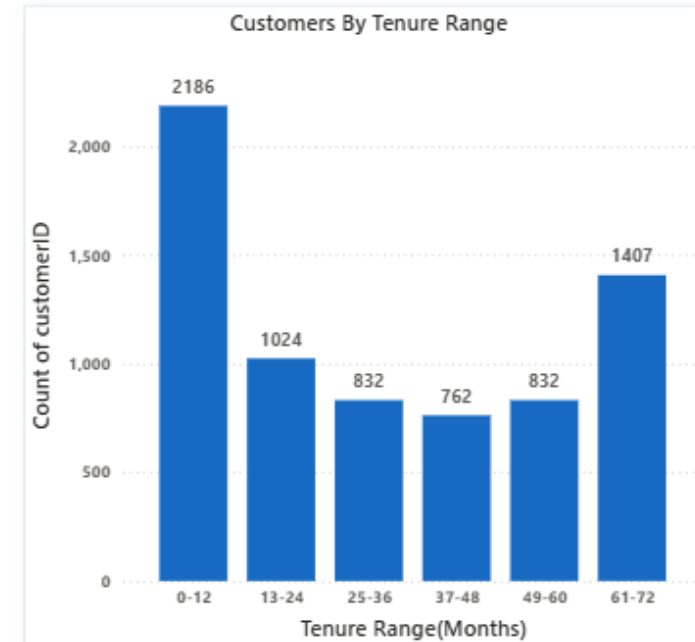


Key Business Insights & Findings

3. How are customers distributed across different tenure ranges?

Key Insight:

- Customers are distributed across six tenure ranges, from 0–12 months to 61–72 months.
- A considerable portion of the customer base is concentrated in the shorter-tenure groups.
- This shows that the company has many relatively new customers compared with customers who have stayed for several years.
- The distribution provides an important base for understanding how churn changes as customer relationships become longer.

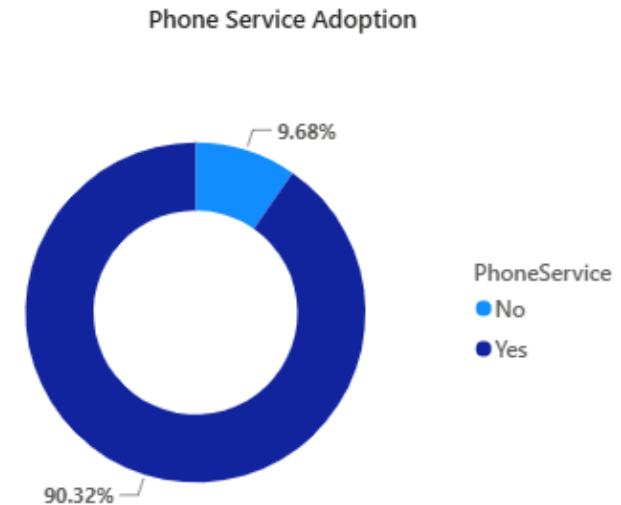


Key Business Insights & Findings

4. How many customers are subscribed to phone service, and how many are not?

Key Insight:

- Most customers in the dataset are subscribed to phone service, while only a relatively small portion do not have phone service.
- This indicates that phone service has already achieved high adoption among the customer base.
- Because most customers fall into the same category, phone service alone provides limited differentiation in the overall customer population.
- Other service characteristics can therefore provide more useful differences when analyzing churn.

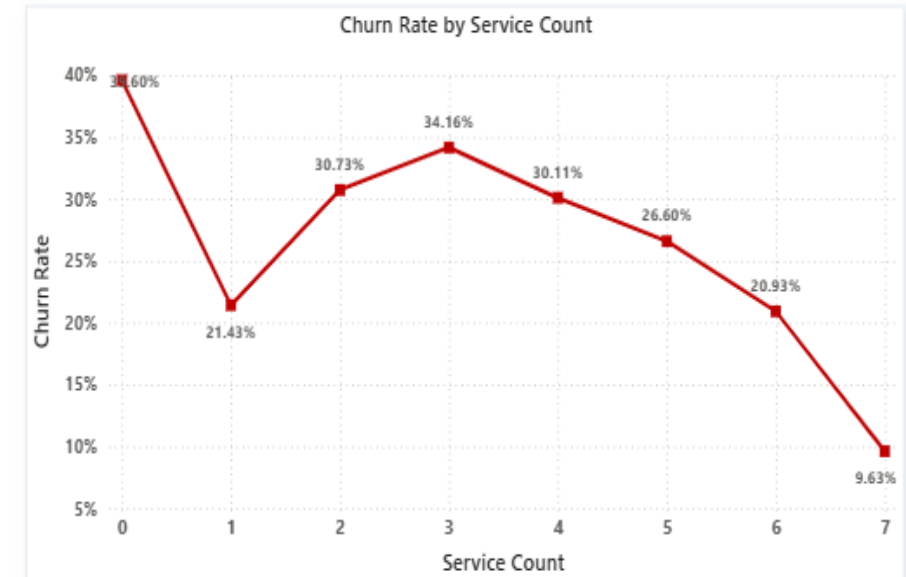


Key Business Insights & Findings

5. Is there a correlation between the number of subscribed services and the likelihood of churn?

Key Insight:

- The churn pattern changes depending on the number of services used by a customer.
- Customers with fewer subscribed services can show a higher concentration of churn compared with customers who have a broader set of services.
- This suggests that customer engagement with multiple services is related to different churn patterns.
- However, service count should be considered together with factors such as tenure and contract type for a more complete analysis.

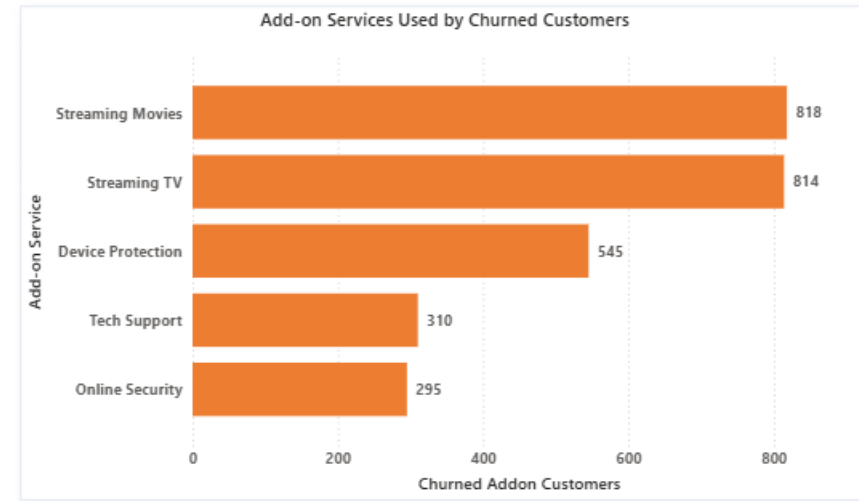


Key Business Insights & Findings

6. Which add-on services are most commonly used by customers who churned?

Key Insight:

- Some add-on services are more commonly used by customers who churned.
- Customers using these services show a higher tendency to leave.
- Other services have lower usage among churned customers.
- The company can focus on these services to improve customer retention.

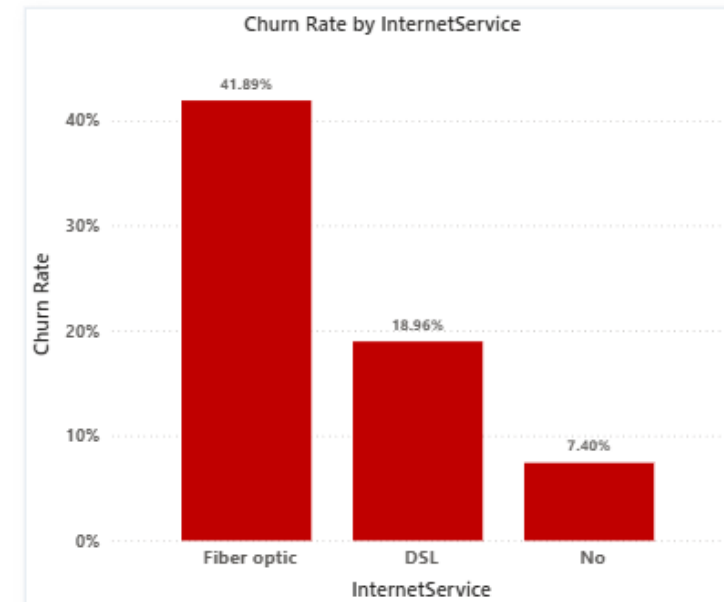


Key Business Insights & Findings

7. How does churn differ between Fiber, DSL and customers without internet?

Key Insight:

- Fiber optic customers stand out with the highest churn rate of 41.89%, while DSL customers have a lower rate of 18.96%.
- Customers without internet service have the lowest churn rate at 7.41%.
- The difference between fiber optic and no-internet customers is particularly large.
- This shows that internet service category is one of the more noticeable differences in the churn analysis.

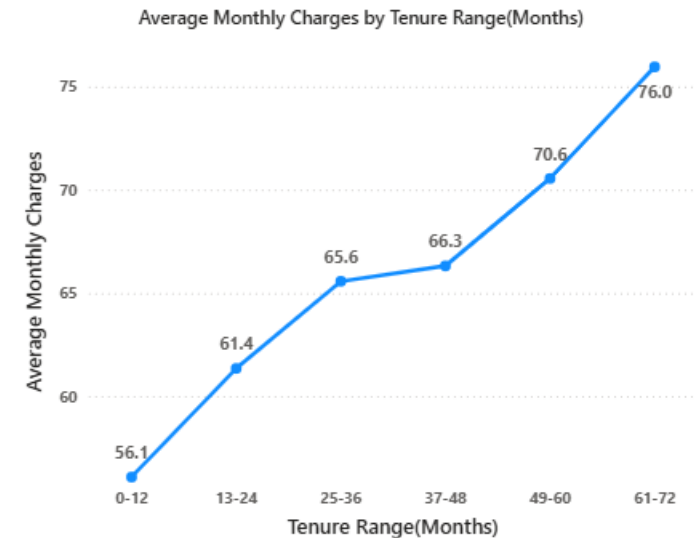


Key Business Insights & Findings

8. Does the average monthly charge increase with higher tenure?

Key Insight:

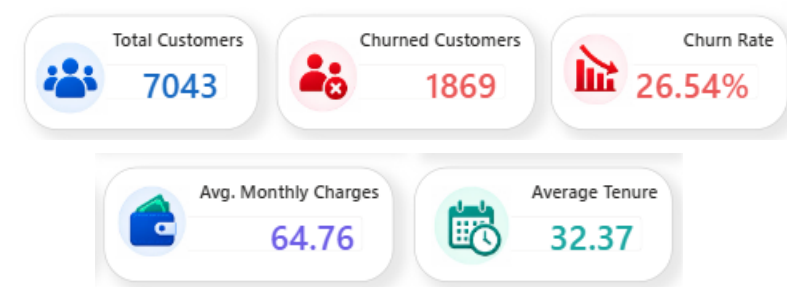
- The average monthly charge generally shows an upward pattern across the higher tenure groups.
- Customers who have stayed with the company for longer periods tend to have higher monthly charges.
- This indicates that longer-tenure customers are not only more likely to remain but can also represent higher-value customer relationships.
- The pattern helps connect customer longevity with monthly revenue contribution.



Key Business Insights & Findings

9. What are the key customer metrics with interactive filters?

- **Key Insight:**
The dashboard brings together important metrics such as total customers, churned customers, churn rate, average monthly charges, and average tenure.
- These metrics can be explored using interactive filters such as gender and contract type.
- This allows the overall customer picture to change dynamically based on the selected segment.
- It makes comparison between different customer groups easier and more informative.

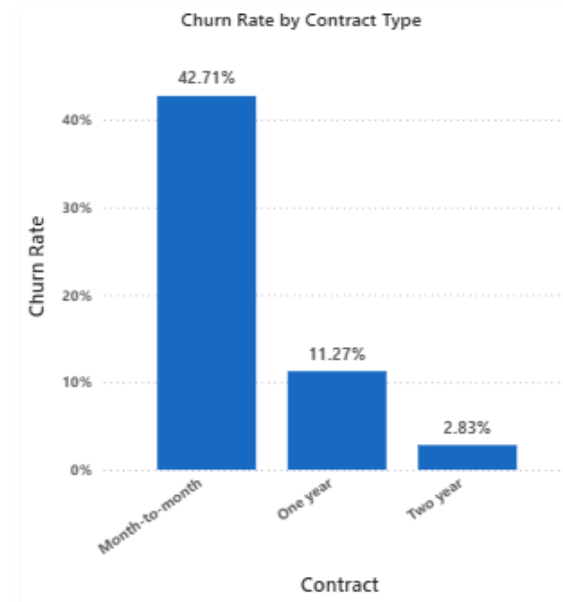


Key Business Insights & Findings

10. How does churn vary across different contract types?

Key Insights:

- Contract type shows one of the clearest differences in customer churn.
- Month-to-month customers have a churn rate of 42.71%, compared with 11.27% for one-year contracts and only 2.83% for two-year contracts.
- The churn rate decreases sharply as the contract period becomes longer.
- This indicates a strong relationship between contract commitment and customer retention in the dataset.



Key Business Insights & Findings

Customer Overview & Demographics

- Churn is highest at 47.44% for 0-12 months tenure and declines with tenure.
- Average monthly charges increases from 56 to 78 as tenure increases
- Customers are almost evenly Split: 50.48% Male and 49.52% Female
- Monthly Charges are similar for males and females across tenure ranges

Service Subscription Analysis

- 90.32% of customers use Phone Service.
- Fiber optic has the highest churn (41.89%).
- Churn peaks at 34.16% for customers with 3 services.
- Streaming Movies and TV are the top add-ons among churned customers.

Contract, Billing & Churn Drivers

- Month-to-month customers have the highest churn rate (42.71%).
- Electronic check users show the highest churn (45.29%).
- Churn resulted in approximately 2.86M lost revenue.
- Month-to-month + Fiber optic customers have the highest churn (54.61%)

Conclusion & Recommendations

Conclusion

- Developed an interactive Power BI dashboard for customer churn analysis.
- Analyzed demographics, services, contracts, and billing.
- Used DAX, KPIs, and visualizations to identify key churn patterns.
- Provides actionable insights to support customer retention.

Recommendations

- Improve early customer engagement and onboarding.
- Promote long-term contracts with loyalty offers.
- Address issues in high-churn services and payment methods.
- Use targeted retention campaigns for high-risk customers.
- Implement predictive churn analysis in the future.

Challenges Faced During project

- Creating meaningful DAX measures such as churn rate, service adoption rate, average services per customer, and retained/lost revenue.
- Avoiding repetition across 3 dashboard pages while covering different aspects of customer churn.
- Choosing suitable visualizations when many charts were initially becoming similar column/bar charts.
- Handling service analysis and correctly calculating the number of services used by each customer.
- Designing the dashboard layout and colors to make all three pages consistent, professional, and easy to understand.
- Creating interactive navigation and filters so users can easily explore the dashboard.

Future improvements

- Predict churn: Identify customers who may leave in the future.
- Risk score: Give each customer a churn risk score.
- Real-time updates: Keep the dashboard updated with the latest customer data.
- Early alerts: Notify the team when a customer has a high churn risk.
- Customer-level analysis: Analyze individual customers to understand why they may leave.
- Better retention: Use these insights to create personalized offers and retention strategies.

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